

# Some GCN-enabled milestones

- **1999 – GRB 990123**

First prompt optical flash detected from a GRB (ROTSE).

GCN enabled near real-time follow-up within seconds of the BATSE alert.

- **2017 – GW170817 / GRB 170817A**

First gravitational-wave and gamma-ray joint detection (LIGO/Virgo + Fermi/INTEGRAL).

GCN distributed coordinated alerts worldwide within minutes.

Marked the birth of multi-messenger astronomy.

- **2017 – IceCube-170922A**

First high-energy neutrino linked to an astrophysical source (blazar TXS 0506+056).

GCN alerts unified neutrino, X-ray, gamma-ray, and optical observations.

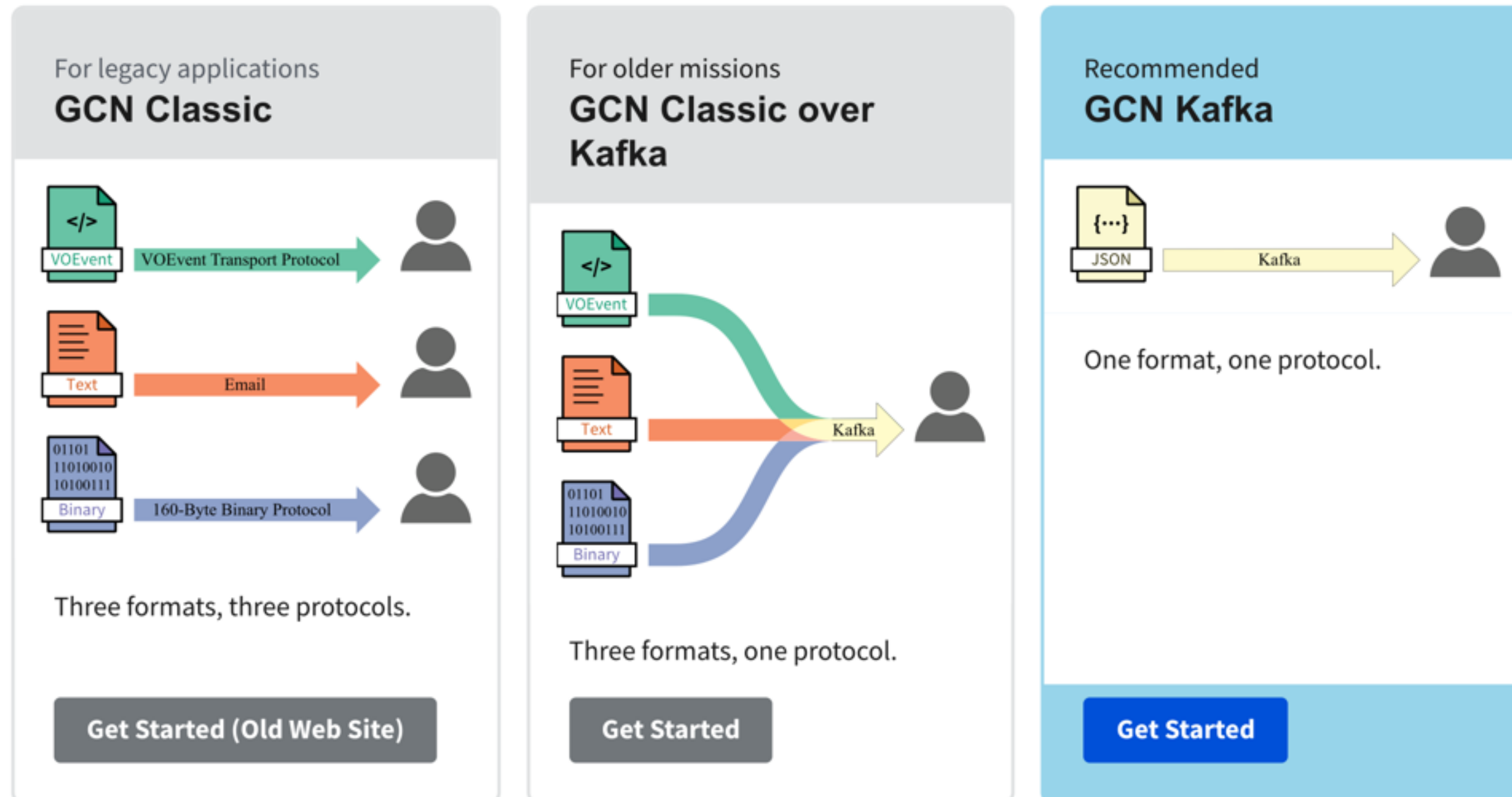
- **2022 – GRB 221009A (“The BOAT”)**

The Brightest Of All Time, detected across the spectrum.

GCN alerts allowed rapid follow-up from Fermi, Swift, INTEGRAL, H.E.S.S., MAGIC, and others.

Showcased the power of modern GCN for real-time global coordination.

# What's new about GCN



A **protocol** defines **how data and alerts are communicated** between systems — that is, the *language, format, and transport method* used to send alerts from detectors to subscribers

## WHY APACHE KAFKA

- **Publish/subscribe model:** producers (missions) publish events, consumers (observers, brokers) subscribe to topics
- **Real-time streaming** with millisecond-scale latency
- **Reliable delivery and persistence** — clients can reconnect and replay missed messages
- **Scalable** — supports hundreds of missions and thousands of subscribers
- **Uniform JSON schema** — no need for mission-specific parsers